

**INTEGRATED CLIMATE-INTELLIGENT SALT
PRODUCTION ECOSYSTEM: A MULTI-MODAL AI
FRAMEWORK FOR SUSTAINABLE YIELD
OPTIMIZATION IN TROPICAL COASTAL
ENVIRONMENTS**

Project ID: 25-26J-431

Project Proposal Report

Ansar TD – IT22893734

Supervised by: Dr. Nathali Silva

Co-supervised by: Mrs. Thilini Jayalath

B.Sc. (Hons) Degree in Information Technology, specializing in
software engineering

Department of Software Engineering
Faculty of Computing

Sri Lanka Institute of Information Technology

Sri Lanka

August 2025

**INTEGRATED CLIMATE-INTELLIGENT SALT
PRODUCTION ECOSYSTEM: A MULTI-MODAL AI
FRAMEWORK FOR SUSTAINABLE YIELD
OPTIMIZATION IN TROPICAL COASTAL
ENVIRONMENTS**

Project ID: 25-26J-431

Project Proposal Report

Ansar TD – IT22893734

Supervised by: Dr. Nathali Silva

Co-supervised by: Mrs. Thilini Jayalath

B.Sc. (Hons) Degree in Information Technology, specializing in
software engineering

Department of Software Engineering
Faculty of Computing

Sri Lanka Institute of Information Technology

Sri Lanka

August 2025

DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student Number	Signature
Ansar T. D	IT22893734	

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the Supervisor: 

Date: 22/08/2025

Signature of the Co-Supervisor: 

Date: 22/08/2025

ABSTRACT

The Intelligent Planning component is the tactical decision-making heart of the Integrated Climate-Intelligent Salt Production Ecosystem to streamline the climate-vulnerable salt industry operations in Sri Lanka. By combining real-time environmental information, market forces, regulatory requirements, and user-specific business objectives, this module develops responsive, scenario-based operational plans that enhance profitability, resilience, and sustainability. As compared to traditional manual planning, it also incorporates superior AI methodologies like rule-based optimization, reinforcement learning, or fuzzy logic models that can dynamically generate and refine actionable planning that closely balances climate impact and market-driven conditions to produce salt.

Development is based on a logical multi-step development process beginning with the requirements analysis and a thorough data mapping of environmental, economic, and operational parameters. The scalable data pipeline ingests a variety of heterogeneous data, such as historical data and market data, to ensure that there is no variance in the timing of data inputs to planning algorithms. With the ability to source, operate, and deliver through cloud computing and modular architecture, the system has predictive and prescriptive insight built into an intuitive dashboard interface that is customizable and is real-time responsive.

Evaluation will be done on historical back-tested planning scenarios, simulation under a range of climate scenarios and market conditions, and pilot with industry stakeholders, providing data to refine iteratively through operations-based feedback. The business value of the system is evident in its ability to minimize harvest risk, adapt to regulatory requirements, and serve disparate salt profiles, including those that represent small cooperatives and those that lean toward large industrial production, plus the system supports the rejuvenation of the industry of salt in Sri Lanka.

The commercialization routes can be bullet-pointed as a tiered Software-as-a-Service (SaaS) subscription scheme, supply chain data services to actors, and perhaps strategic alliances with governments and development organizations. Its multi-language and cloud-based structure will enable scale-up to other tropical salt-producing areas with similar environmental issues at a regional and global level. The Intelligent Planning component is an example of combining the latest AI with domain expertise in transforming traditional salt farming into a data-driven, climate-resilient sector to underpin economic livelihoods and nation-state self-sufficiency, as climate risk evolves.

TABLE OF CONTENTS

<i>DECLARATION</i>	<i>iii</i>
<i>ABSTRACT</i>	<i>iv</i>
<i>TABLE OF CONTENTS</i>	<i>v</i>
<i>LIST OF FIGURES</i>	<i>vii</i>
<i>LIST OF TABLES</i>	<i>viii</i>
<i>LIST OF ABBREVIATIONS</i>	<i>ix</i>
<i>1. INTRODUCTION</i>	<i>1</i>
<i>1.1. BACKGROUND AND LITERATURE SURVEY</i>	<i>2</i>
<i>1.2. RESEARCH GAP</i>	<i>3</i>
<i>1.3. RESEARCH PROBLEM</i>	<i>5</i>
<i>2. RESEARCH OBJECTIVES</i>	<i>7</i>
<i>2.1. MAIN OBJECTIVE</i>	<i>7</i>
<i>2.2. SPECIFIC OBJECTIVES</i>	<i>7</i>
<i>3. METHODOLOGY</i>	<i>9</i>
<i>3.1. System Overview</i>	<i>9</i>
<i>3.2. Individual Component</i>	<i>11</i>
<i>3.2.1. Requirement Gathering and System Design</i>	<i>11</i>
<i>3.2.2. Development of the Web Platform</i>	<i>12</i>
<i>3.2.3. Testing and Quality Assurance</i>	<i>13</i>
<i>3.2.4. Data Collection and Analysis</i>	<i>13</i>
<i>3.3. Materials Required to Carry Out the Project</i>	<i>14</i>
<i>3.3.1. Equipment</i>	<i>14</i>
<i>3.3.2. Software and Computer Support</i>	<i>14</i>
<i>3.3.3. Other Resources</i>	<i>15</i>
<i>3.4. Data Required for the Project and Collection Methods</i>	<i>16</i>
<i>3.5. Estimated Time Frame</i>	<i>18</i>

3.6.	<i>Work Breakdown Structure</i>	18
3.7.	<i>Commercialization</i>	19
3.7.1.	<i>Product Differentiation and Value Proposition</i>	19
3.7.2.	<i>Revenue Model</i>	19
3.7.3.	<i>Scalability and Market Penetration</i>	20
3.7.4.	<i>Marketing Strategy</i>	20
4.	PROJECT REQUIREMENTS	21
4.1.	<i>Functional Requirements</i>	21
4.2.	<i>Non-Functional Requirements</i>	22
4.3.	<i>Test Cases</i>	22
4.4.	<i>Wireframe</i>	23
5.	BUDGET AND BUDGET JUSTIFICATION	24
6.	REFERENCES	25
7.	APPENDICES	27

LIST OF FIGURES

Figure 1.1.1: Comparative Analysis of Existing Systems vs. Proposed Solution.....	5
Figure 1.3.1: Diagram to represent Research Gap	6
Figure 3.1.1: System Diagram	10
Figure 3.5.1: Work Breakdown Structure	18
Figure 4.4.1: Wireframe For Intelligent Planning Dashboard.....	23
Figure 4.4.2: Wireframe for Scenario Simulation Dashboard.....	23

LIST OF TABLES

Table 0.1: List of Abbreviations	ix
Table 3.5.1: Estimation Time Plan	18
Table 4.3.1: Test Cases	22

LIST OF ABBREVIATIONS

Table 0.1: List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
DaaS	Data-as-a-Service
ERP	Enterprise Resource Planning
GPU	Graphics Processing Unit
IoT	Internet of Things
LSTM	Long Short-Term Memory
ML	Machine Learning
RMSE	Root Mean Square Error
SaaS	Software-as-a-Service
UI	User Interface
AWS	Amazon Web Services
GCP	Google Cloud Platform
ADB	Asian Development Bank
UNDP	United Nations Development Programme
ML	Machine Learning

1. INTRODUCTION

The Puttalam salterns in the northwest of Sri Lanka, long regarded as the mainstays of the national salt industry, have depended on generations on large shallow ponds and traditional methods of solar evaporation. Although these methods have in the past been effective and environmental-friendly in a stable and dry environment, the operations of the industry have been undermined in recent years by the increasing climate variability, greatly affecting predictability and sustainability. It is noteworthy that the Sri Lankan salt production decreased by more than 40 percent in 2023-2024 because of heavy and unseasonal rains exceeding the average of rainfall recorded in the past (1,895 mm in 2024 in saltern districts); this was unprecedented [1]. This has led to the Ministry of Industries tendering to import 30,000 metric tonnes of non-iodized salt in early 2025 as a direct result of meeting domestic shortages. These upheavals subsequently spill into the local economy- disrupting employment, eroding export potential and casting doubt on national food and industrial security.

Recent Intergovernmental Panel on Climate Change (IPCC) reports affirm that global warming has been increasing the frequency of flooding, disruptive rain patterns, a rise in sea level, and more severe storms in tropical nations such as Sri Lanka, with direct effects on the salt-producing economies of such nations, which have little adaptive capacity. [2] Directors of salt companies and local saltern officers have been observing that due to new rain events being unpredictable and fierce, the ponds have continued to be diluted, with endless dissolving of new crystals of salt and cancelling harvests over long intervals. This context of change is therefore no longer sustainable in terms of empirical knowledge and manual fragmented management.

AI and ML are transforming agriculture and industry around the world with predictive analytics, automated quality inspection, and distributed, privacy-conscious optimization-summons now being applied to crop planning, fisheries quota forecasting, and on-the-fly supply chain reconfiguration. As an example, in other countries, commercial AI-based forecasting models like Salt Bae have proven effective, and IBM Watson Agriculture is widely used to do adaptive crop scheduling. Nonetheless, such technologies have not been scaled and completely adopted at Puttalam and Hambantota salterns, even though there are pilot demonstrations. The operators mostly use logbook records, which are decades old, personal weather forecasts, and market intermediaries' indications on the prices. A lack of real-time, multimodal planning tools and sophisticated data privacy measures (a key competitive advantage of cooperatives) would signify an important novelty and research need in ag tech in Sri Lanka. [3]

On past interventions in the salt industry, emphasis has been laid on physical upgrades (mechanization, better filtration), but emphasis has not been laid on the integration of the environmental, operational, and market data streams. The proposed Integrated Climate-Intelligent Salt Production Ecosystem fills this gap with a digital platform that brings together forecasting, adaptive planning, real-time computer vision quality control, and federated learning to optimize secure and distributed waste management. Particularly, the implementation of federated learning is innovative in this regard: unlike traditional cloud-based applications, federated learning in the context of Sri Lankan cooperative salterns facilitates the collaborative training of AI models on local and privately controlled data sets without direct data transfer,

significantly improving privacy and regulatory compliance - a practice that has not been operationalised within ag-tech technology very often, but is now starting to be explored in distributed palm oil and precision cropping projects.

In this ecosystem, the Adaptive Intelligent Planning Engine is a transformation engine. It integrates real-time meteorological streaming, ground-level production reporting, dynamic regulations, and user-defined business goals, allowing both operational resiliency in the short term and profitability in the long term in an environment that is becoming climate sensitive, such as Puttalam. This transformation-reactive and tradition-based model to a proactive and AI-focused paradigm will underpin the long-term social, economic, and environmental gains, with Sri Lanka leading the region in its climate-adaptive salt production.

1.1. BACKGROUND AND LITERATURE SURVEY

One of the most ancient industrial processes is the extraction of salt using the method of solar evaporation, which is traditionally observed in the tropical coastal settings, since high temperatures and scarcity of rainfall establish an environment in which salt can easily crystallize [4] Sri Lanka has major salterns in Hambantota, Puttalam, and Elephant Pass that plays a significant role in supplying salt in the country [5] However, this has changed drastically over the past few years because of seasonal anomalies and heightened climate variability, which has complicated the conventional production cycles making them adapt to the operations without the use of effective predictive and planning tools [6]

Present operational planning in these salterns relies mostly on manual estimation and past averages usually become inadequate in unpredictable weather conditions. The absence of integrated decision-support systems makes operators provide crucial decisions based on incomplete information (weather predictions of publicly funded weather agencies, market prices provided by traders, and informal estimates of yield using subjective judgment of experience), leading to inefficiencies and lost reliability in production [7].

The use of artificial intelligence (AI) in production planning has proved to be successful in other countries that are in close reliefs in terms of resource quest. As such, in agriculture, AI can be used to plan crops adaptively based on weather, soil, and market data to maximise yields [8] . Predictive analytics are used in the fisheries industry, in which it runs on maximizing fish quotas to operate in an unstable environment [9] . Likewise, in manufacturing, the reinforcement learning algorithm is utilized in improving resource scheduling and operational prowess. [10] Such global progress notwithstanding, these AI-driven methodologies have not effectively been modified to deal with the unique interaction of the environmental, economic, and regulatory complexities that exist in the salt industry in developing nations such as Sri Lanka.

Sri Lanka has a salt industry facing two kinds of pressure, which are imposed on it by the market dynamics and by the regulatory frameworks. During domestic trading, the spheres need stable plans and cheap salt, and in communities abroad, traders must meet strict purity standards by international agencies [11]. A respectable operational planning system has to wade through these perhaps competing objectives without compromising on economy as well as respecting environmental sustainability requirements [12]

In this literature, there is an urgency to use integrated data-driven decision-support solutions involving climatic forecasts and market trends, as well as regulatory requirements that will facilitate the more proactive and efficient operational planning in Sri Lankan salterns, and most importantly in Puttalam. This growth of such systems holds out the promise of enhancing adaptation to climate variability, maximization of resource utilization as well as quality of products and hence will tackle problems around the globe and locally affecting the industry.

1.2. RESEARCH GAP

It has been seen that even though many countries in the world have developed and implemented AI-driven operational planning within an industry like agriculture, fisheries, and manufacturing, an industry-specific and integrated system with regards to salt production is not found in Sri Lanka. Current digital applications in adjacent industries are concerned either with simple weather notification or narrow areas of operation, rather than building a more complete, multi-parameter decision-making regime, specific to salt production.

The key research gaps identified and addressed by the components of the proposed Integrated Climate-Intelligent Salt Production Ecosystem are:

- **The absence of Multi-Parameter Integration:**
Current approaches to planning in the salt industry do not integrate its main inputs such as climate predictions, changing production capabilities, demand levels, operational expenses, and legal requirements. This shortcoming is filled by Component 1: Crystallization Forecasting, which runs on multimodal data crystallization forecasting that incorporates weather forecasts and other measurements and takes the form of highly accurate data-driven forecasts of optimal crystallization conditions and crop harvesting time.
- **Reactive, as opposed to Predictive Decision-Making:**
Existing methods only make decisions based on the observed environmental changes and not on the ability to predict the changes, which results in a loss of efficiency. This gap is filled by component 2, the Adaptive Intelligent Planning Engine (Intelligent Planning Engine) that combines forecasts, market trends, cost

structures, and regulatory requirements into frame-based predictive operational plans that make proactive management proactive.

- **Failure to Adapt in Real Time to New Conditions:**

The market conditions and salt production environments are very unstable and quick to change with regard to weather and demand. Nonetheless, existing systems do not have the capability of automatically varying plans on an almost real-time basis. This obstacle is circumvented by Component 3, which is Computer Vision Quality Control that facilitates real-time automated monitoring of salt quality feeding real-time feedback into operational changes, and Component 2, continual refreshes of planning strategies using real-time input data.

- **Lack of planning tools relevant to the stakeholders:**

The business strategy of saltern operators differs; the owners of one saltern are oriented to the production of high-quality salt that covers export needs whereas others want to increase their output volume to sell it into local markets. Nowadays, there are no tools with such possibility. The user input and customization layer (Component 2) builds into the Adaptive Intelligent Planning Engine a customized level of operation strategies in line with the specific priorities of individual stakeholders.

- **AI Techniques That Are Not Utilized to the Full Potential in Salt Production Intrigues:**

Though AI technologies like reinforcement learning, fuzzy logic, and heuristic optimization are well-developed in other domains, they are not applied to the conditions and the possibilities of salt production in the tropics. The system, where all components are integrated (namely, Component 4: Federated Learning Waste Valorisation), utilizes state-of-practice AI solutions, including federated learning algorithms, to make it possible to coordinate the optimization processes across numerous locations, all the time protecting the privacy of data.

Comparative Analysis of Existing System vs. Proposed Solution

System	Features	Limitations	Our Solution: Improvements
Salt Bae	AI-based salt yield predictions, mobile UI	No regulatory/market integration, lacks privacy frameworks	Compliance, local privacy, regulatory integration
IBM Watson Agriculture	Crop planning, weather analytics, global	Generic; not adapted for salt or tropical microclimates	Salt-specific, local climate and regulation aware

System	Features	Limitations	Our Solution: Improvements
Fisheries AI (Catch Quotas)	Predictive analytics for harvest limits	Marine focus; not for stationary evaporative ponds	Designed for pond production and land processes
Local Manual Practices	Operator experience, logbooks	Lacks data, non-predictive, high risk with climate change	Real-time, predictive, not reliant on experience
Palm Oil FL Platforms	Federated learning for privacy, edge AI	Not applied to salt, lacks market/regs integration	Privacy, plus market/regulation, salt focus

Figure 1.2.1: Comparative Analysis of Existing Systems vs. Proposed Solution

Coordinating these interconnected gaps through a modular yet integrated ecosystem, this project aims at changing how the Sri Lankan salt industry, specifically the Puttalam salterns, is currently operated, due to the current fragmented and reactive management it is based upon, towards sustainable, data-driven, and resilient operational planning.

1.3. RESEARCH PROBLEM

The key research question that will prove to be the impetus of this venture can be put in the following words:

“How would an adaptive, learnt planning algorithm be formulated to incorporate real-time environmental, production, economic factors, and regulatory factors, in the context of generating optimal operational strategies to enhance profitability, efficiencies, and make the salt production industry of Sri Lanka import-dependent with regards to climatic variability?”

This problem statement captures both the technical challenge - creating and deploying an effective, multi-parameter optimization tool that can address complex and dynamic inputs and the practical need, giving salt producers access to sophisticated decision-making tools to respond to an increasingly erratic climate and market environment.

The issue can even be further broken down into three critical dimensions:

- **Environmental Uncertainty:** Dealing with unpredictable changes in weather and climate conditions that affect the salt crystallization process and harvest directly and are largely dependent on changes in temperature, rainfall, and other weather-related conditions.

- **Economic Variability:** Adjusting the production schedule and resource allocation to varying and changing market demand, availability, and price signals in a way that can maximize revenue and minimize wastage and overproduction.
- **Regulatory Compliance:** This is necessary to ensure that operational plans and final products are acceptable as per the strict national and international quality standards and environmental regulations, which are vital in sustaining export market capability and industry sustainability practices.

Solving this complex issue will not only increase the level of self-reliance of salt production in Sri Lanka but also help in increasing the competitive ability of the industry in the local and international markets. Moreover, the plan of developing the solution as a scalable and modular framework will facilitate transferability of the project to other tropical coastal areas of the world, which might be similar to those of the project areas in terms of environmental and operational similarities in salt production.

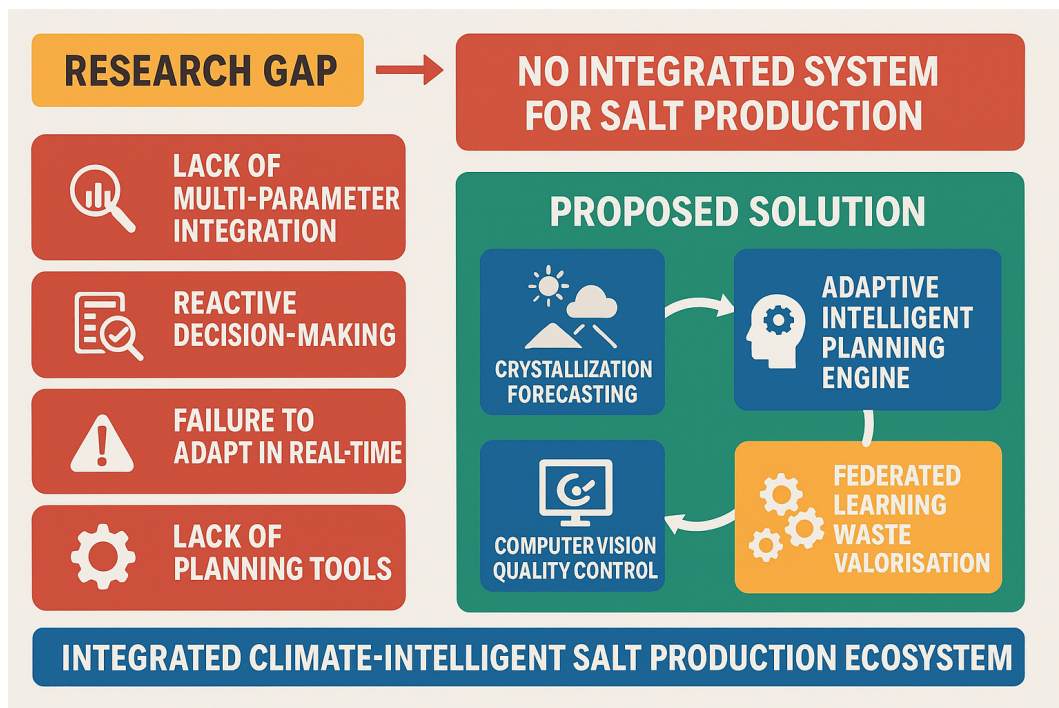


Figure 1.3.1: Diagram to represent Research Gap

2. RESEARCH OBJECTIVES

This section introduces the overall aims of the Integrated Climate-Intelligent Salt Production Ecosystem as a multi-component platform to modernize, optimize and future-proof climate-sensitive coastal salt production in Sri Lanka.

2.1. MAIN OBJECTIVE

In producing and deploying a bespoke, AI-powered ecosystem designed to support the salt production industry in Sri Lanka by facilitating predictive environment analytics, adaptive operating program code, autonomous quality control, and joint optimization of operations resources, which works toward increased productivity, sustainability, and climate adaptation, along the entire salt production value chain.

SMART Breakdown:

- **Specific:** Design and implement an integrated platform that would consist of four autonomous but interconnected AI-powered modules (crystallization forecasting, intelligent planning, computer vision quality control, federated learning waste valorization).
- **Measurable:** The performance of the success is measured through increases in production yield ($\geq X\%$), use efficiency in resources ($\geq Y\%$), quality product compliance ($\geq Z\%$), and the decrease of wastes as well as positive feedback on at least two operational pilots.
- **Achievable:** Achievable by leveraging the project team's expertise in AI, software engineering, and salt industry operations, and by collaborating with local saltern partners for piloting and validation.
- **Realistic:** Realistic to the available technology and ability to access experienced personnel in operation, besides high stakeholder involvement in the industry.
- **Time-bound:** To get fully implemented, deployed, and reviewed even on a project level.

2.2. SPECIFIC OBJECTIVES

The specific objectives below specify the nature and extent of the contributions of each ecosystem component as well as promoting synergy among the components:

- **Crystallization Forecasting:** To develop and validate machine learning models of weather, sensor, and historical production data, and give early and precise deep forecasts of optimal crystallization and harvest windows.
- **Intelligent Planning Engine:** Develop flexible AI-based planning systems that can take predictive forecasts, real-time market, manufacturing, and regulatory information, and convert it into stakeholder-optimized, actionable operational plans to make profitable, proactive decisions.

- **Computer Vision Quality Control:** Implement a high-throughput automated computer vision system to grade the salt and detect salt impurities in real-time so that the high-quality and regulatory standards of the product can be achieved, and any human touch and production bottlenecks are reduced.
- **Federated Learning Waste Valorization:** Develop a federated learning infrastructure, a secure and collaborative setting to optimize the reuse and valorization of salt waste in a distributed set of sites, without compromising data that is sensitive but also enhancing a sustainable circular and economic enrichment initiative.

In attaining these goals, the project will not only assist Sri Lanka in the self-sufficiency in salt production, competitiveness in both regional and international markets, and sustainable and technology-driven management of its resources, but also helping provide the country with the necessary leadership in the above aspects. The replicability of the ecosystem in terms of modularity and scalability also allows the ecosystem to be implemented in other tropical coastal nations that are grappling with the same challenges and thereby making more contributions to the topic of climate-smart industry transformation.

3. METHODOLOGY

3.1. System Overview

The Integrated Climate-Intelligent Salt Production Ecosystem is a modular, AI-driven platform (potentially modular) that aims to revolutionize the salt production industry in Sri Lanka by performing end-to-end decision support tasks to the entire salt production lifecycle. The structure of its architecture is divided into four related components that take care of each of the vital steps in operational planning, execution, and optimization.

System Components

1. **Crystallization Forecasting:** Combines multimodal information (weather, other) with advanced time series models (ARIMA, Prophet, and LSTM) to determine the ideal conditions to crystallize salt and harvest it. This makes it possible to develop proactive plans and to reduce the risks of yields depending on the uncertainty of the climate.
2. **Intelligent Planning Algorithm:** Converts predictive intelligence of Component 1, with real-time market trends, production capacities, economic data, and regulatory requirements into optimized operating strategies implemented into actions. It uses multi-parameter optimization, heuristics, fuzzy logic, decision trees, and reinforcement learning to support dynamic data-driven planning, which can change as circumstances change.
3. **Quality Control: Computer Vision:** Automatically grades the salt quality using computer vision image analysis and impurities are detected in real time. This improves management without the need of manual check-up, thereby making it more regulated and assuring the quality of the products to both the local and international markets.
4. **Federated Learning Waste Valorization:** Facilitates the distributed and privacy-preserving coordination of different sites of the saltern to optimize by-product reuse and valorization. By using federated learning and differential privacy, they can be sure that critical operational data will not be shared directly, but it will lead to innovation and data safety.

Key Features

- Modularity
- Real-time Data
- User-Centric Dashboard

Such an interconnected ecosystem has the unique opportunity to consider the complex climate sensitive operation realities of the salt industry in Sri Lanka.

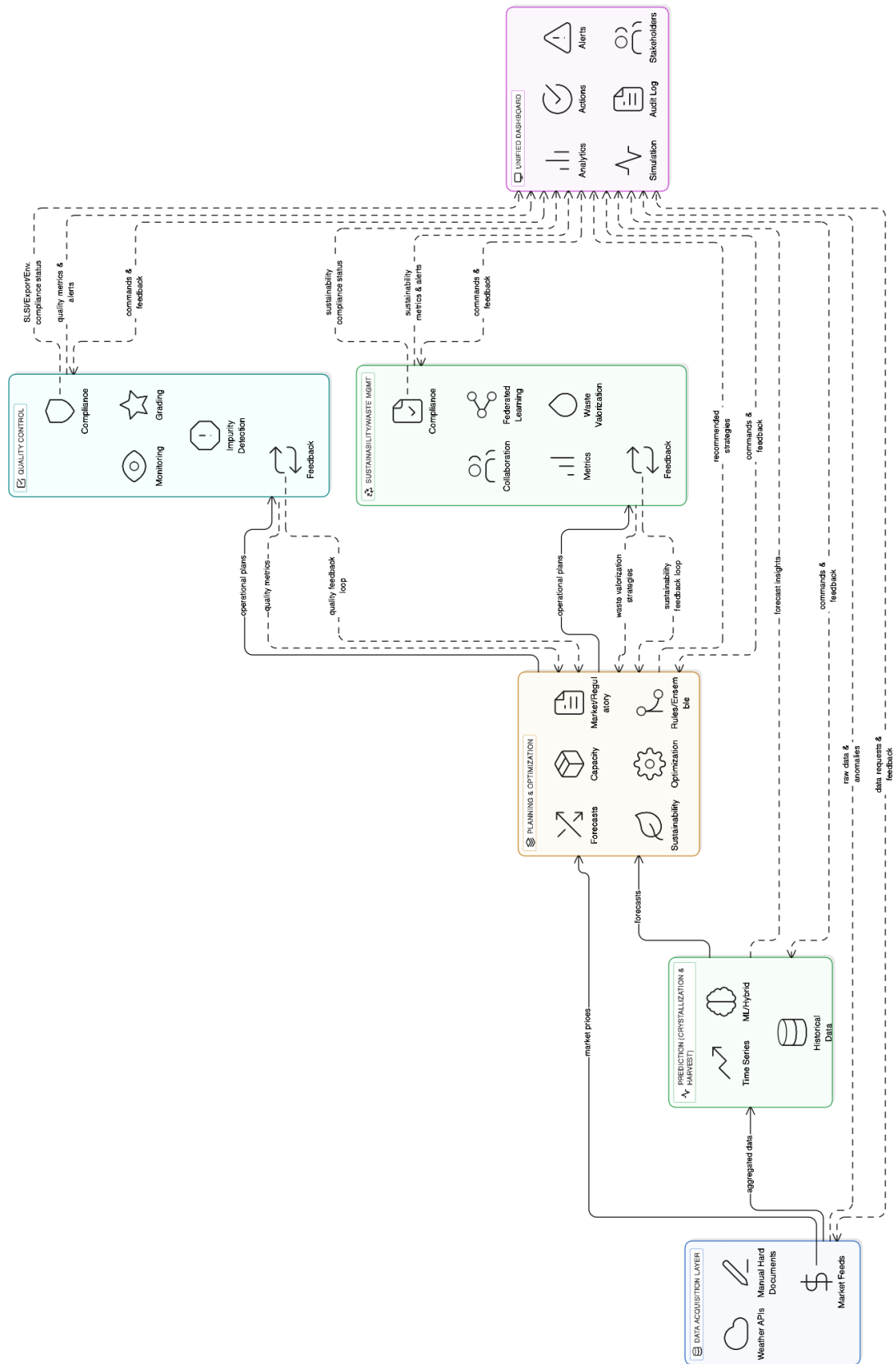


Figure 3.1.1: System Diagram

3.2. Individual Component

The algorithm of the Intelligent Planning is the brain of the Integrated Climate-Intelligent Salt Production Ecosystem, conceptualized as decision-making. It was the key connecting factor between predictive analytics, operational information, stakeholder preferences, and regulatory standards to allow the salt industry in Sri Lanka to shift to proactive, data-driven, and optimized production planning operations rather than the reactive, manual, and fragmented production planning. This element integrates the contributions of environmental forecasts, operational and real-time production data, economic and regulatory needs, and produces actionable plans that are most effective, profitable, legally compliant, and sustainable in the long term, even during times of extreme weather and market fluctuations.

3.2.1. Requirement Gathering and System Design

The development begins with the comprehensive stage of requirement gathering and parameter definition because the richness of the algorithm and its reliability is determined by the completeness of the input provided and the artificial clarity of the structure. During this phase, interviews and workshops shall be conducted with the industry stakeholders such as the saltern managers, the line workers, the market analyst, the regulatory officials, and the quality controls, to identify all operating, financial, environmental, and compliance variables that affect the production planning.

- **Environmental Parameters:** IPA consumes the results of the crystallization forecasts and data available in Component 1 about weather details and historical and current readings of temperature, humidity, rainfall, wind, and salinity of ponds. They are acquired by local weather diaries and manual logbooks, which are digitized and fed into the system.
- **Production Parameters:** The planning tool incorporates the current and the past data on pond capacity, employee posting, seasonal labor fluctuation, machine utilization, upkeep procedures, and downtimes. Information is gathered through the production logbook, resource allocation logs, and maintenance logs.
- **Economic Parameters:** Sales records and interviews with the traders extract monthly and annual sales ledgers, the price list, customer type (e.g., domestic/export/industry), and the demand distribution patterns, which change. The rest of the factors, labor cost, equipment cost, transport cost, and utility cost, are extracted from the accounting and business management records.
- **Quality Parameters:** Desires as to product grade distribution (labels domestic, export, industrial) and returns on grading of Component 3 (e.g., grade rejections, defect levels, appearance standards) keep the system in market and customer harmony.

- **Regulatory Parameters:** The regulatory constraints are based on the compliance requirements (SLSI limits, environmental release, and workplace health requirements). Formal regulation reports of inspection, and communication with compliance officers are the sources to obtain updates.

Data collected is verified (that is, cross-verified across sources), when necessary translated (from Sinhala/Tamil), and scanned with the high-resolution imaging system and standardized to a uniform scientific unit that can allow easy AI modeling. Relationships between these parameters are modelled, thereby providing a clear, logical and time-proof data basis to all subsequent planning decisions.

3.2.2. Development of the Web Platform

The Intelligent Planning Engine will be implemented as a web-based platform that will enable the timely availability of decision support to various parties. Major highlights are:

- **User Interface:** Interactive dashboard with full functionality for saltern managers, owners and planners. It graphs suggested schedules of production, sales plans according to the market, the status of regulations of compliance, and environmental sustainability stats.
- **Customization Functionality:** Able to input and change business priorities (profit, quality, sustainability), market focus (domestic vs. export), and the level of risk tolerance, which dynamically affects the weighting of optimization within the algorithm.
- **Data Integration:** APIs will be consumed to bring in continuous data feeds such as market prices, weather updates, and production logs. It assists in manual data entry of contextual information that is not automatically configured.
- **Scenario Analysis Tools:** Allows stakeholders to conduct forces of nature scenario analyses simulating what would happen to the business in the event of a sudden price adjustment, bad weather, or labor disruptions, and can be used to preemptively manage risks.
- **Security and Accessibility:** Designed on a secure web platform using role-based access control to secure sensitive operational and business data or information.

Such an architecture of the platform allows this module to work fluidly with the rest of the ecosystem (forecasting, quality control, federated learning) to effectively exchange data and elevate control over operations.

3.2.3. Testing and Quality Assurance

Quality, safety, and control testing policies form the basis of a trustworthy and successful IPA system:

- **Unit and Integration Testing:** Data ingestion, optimization engine, and user interface are examples of algorithmic subcomponents, which will be tested in isolation and end-to-end to ensure stability and correctness are being stressed.
- **Simulation-Based Validation:** Simulation-based testing is run on historical production, market, environmental, and quality data. The platform is evaluated on the precision of the operational plans, profitability upsides, and regulatory compliance penetrations under simulation conditions.
- **Pilot Deployments:** Live testing with partners at Puttalam saltern in an iterative way measures real-world performance. Their operational data, user experiences, and compliance results are monitored constantly so as to identify any deficiencies and inconsistencies compared to simulation forecasts.
- **User Feedback Mechanisms:** Operator Feedback via structured questionnaires, interviews, and observation sessions will be obtained, as well as usability ratings to be used in subsequent iterations to improve the interface and algorithm.
- **Performance Metrics:** There are proxy performance measures, including plan compliance rate, forecast compliance, waste reduction, profit margin improvement, and regulatory compliance instances, to spur quality performance.

3.2.4. Data Collection and Analysis

The soundness of the decision-making performed by the IPA is closely linked to the thoroughness and quality of the data on which it is based. Data collection activities work on:

- **Business & Market Data:** The frequencies of sales volumes, price adjustments, and the customer segmentations have been collected regularly, along with the seasonal trends of demand, to balance the realism of the market fluctuations.
- **Operational Planning Data:** Expressive records of manpower distribution (by pond and seasonal exercise), utilizing equipment and maintenance programming, and course of operation judgment format (harvest freighting, market hitting goals).
- **Financial Cost Information:** Season-based labor costs, costs of maintenance, transport, energy, or utility use.
- **Decision-Making Patterns:** Qualitative evidence on the way in which the historical operational decisions are taken and how quality versus quantity trade-offs are considered was recorded in documents and through the interviews.

The data digitization schemes would have involved photographing the handwritten logbooks, translation of Sinhala and Tamil records, converting to double-check records

by multiple sources, and standardization of units to internationally known measures to ensure the data is reliable and can be interoperated.

Through analysis pipelines, data processing is ready before clean ingestion by the IPA, with real-time data flow allowing recalculation of operational plans on the fly, key capabilities reflecting both the dynamic and adaptive properties of the system.

3.3. Materials Required to Carry Out the Project

The performance of the Intelligent Planning Algorithm (IPA) of the Integrated Climate-Intelligent Salt Production Ecosystem operation and subsequent commercialization depends on the presence, compatibility, and interaction of the strong hardware, intelligent software stacks, cloud facilities, data flows, and work coordination tools. Different types of materials and resources are essential to make the system research-grade, scalable as well as resilient to operate.

3.3.1. Equipment

Development Workstations: Very powerful laptops (or desktop computers) are important to code development, model training, code testing, and simulation. The workstations must have at least 16GB of RAM, a multi-core processor (quad-core, etc.), so the machine learning operations can be completed faster and enable parallelism to train and validate models.

Network Infrastructure: High-bandwidth, reliable internet connectivity is paramount to providing the people on site, cloud services, and remotely located development teams a smooth experience of communicating in real time. This consists of routers, switch, and cellular failover as a way of avoiding downtime. Satellite or reserve wireless connections can be taken into consideration in instances where regional coverage cannot be relied upon to ensure completeness of data.

3.3.2. Software and Computer Support

- **Programming Languages:**

Python: The first option in the case of the development of AI/ML models, prototyping algorithms, and backend data pipelines because of the highly advanced library ecosystem and strong communal perspectives. **JavaScript/TypeScript:**

JavaScript or TypeScript is suitable for implementing interactivity and animation in the dashboard and frontend user interface layer, so that end-users are provided with seamless, browser-based experiences.

- **Frameworks and libraries:**

TensorFlow, PyTorch, scikit-learn: These libraries can be used to create, train, and deploy any AI or machine learning model (such as neural networks, down to decision trees, reinforcement learning agents, and hybrid solutions).

Flask/FastAPI: The lightweight (fast) backend frameworks to build REST APIs, which allow establishing the connection between the web interface, database, and the planning algorithm.

React.js/Next.js: Established frontend libraries/ frameworks to create an intuitive and responsive dashboard and management tools of the saltern stakeholders.

- Database Systems:

PostgreSQL: a powerful relational database supporting structured business, operational and compliance data.

MongoDB: Schema-free, free-structured data to store sensor logs, quality image and user-input files.

- Version Control:

GitHub/GitLab: Distributed source control systems which facilitate collaborative code development, branching, issue management, and continuous integration/deployment processes (CI/CD) that are essential to team-based software engineering projects.

- Data visualization tools:

plotly, Chart.js, D3.js: libraries/ frameworks that enable advanced data visualization and data analysis in the dashboard, converting big, multidimensional datasets into readily analyzable, clear outcomes.

- Cloud Services:

AWS/Azure/GCP (Google cloud platform): The providers offer scalable compute resources to host applications, store data, train inference models, and deployment, security and backup services are offered on a managed basis. Third-party data API integrations and disaster recovery planning is presented by cloud platforms as well.

3.3.3. Other Resources

Market Data APIs: Real-time optimization and planning flexibility through the access of near real-time APIs that are frequently updated across salt pricing, and demand information coupled with commodity market patterns. Such may be the industry specific feeds, the government export/import dashboards or the contracted third party analytics vendors.

Regulatory Documentation: Concise and up to date information regarding SLSI salt standards, food safety procedures, environmental regulations and documentation as well as trade/export regulations should be available to facilitate coding of regulatory limitations and cover all plans within the scope of the law.

3.4. Data Required for the Project and Collection Methods

The key to the capability of the Intelligent Planning Algorithm to optimize salt production and to respond to changes in the market, operational and regulatory realities in the coastal salterns of Sri Lanka is a rich and correct dataset. The next section and procedures are targeted to perform an immediate load bearing of business and operations, as well as planning functionality of this Component.

Business & Market Data

- **Sales Records:** Sales volumes and price of a product through the month and years, price differences in accordance with duration and grade of products, customer (local, export, industrial) and seasonal changes in demand.

Collection Technique: Collected on the basis of financial sales ledgers maintained by saltern owners as well as sales managers and local traders.

- **Cost Data:** Separated expenses of labor by the season, equipment maintenance costs, transportation costs and utility overheads (when there are).

Collection Method: Appropriated through accountants records, intern inner financial records and consultations with operational managers.

- **Market Timing Insights:** Information about when and why sales are undertaken, how decisions about market entry are achieved, and the current quality/quantity trade-offs.

Mode of Collection: TheSqliles of sale managers, local buyers, and traders
Structured interviews.

Data of Operational Planning

- **Resource Allocation Records:** Employee allocation records by pond/production area, equipment utilization and frequent maintenance schedules and records of seasonal workforce fluctuations.

Collection Method: References: Get to know operational logbooks, staff schedules, and maintenance records as kept by production managers or supervisors.

- **Decision-Making Patterns:** Accounts and procedural explanations of how harvesting timing decisions are reached, market match decisions, grade setting policies, and other planning-key operational decisions.

Collection Method: Face-to-face interviews and group discussions with the saltern owners, managers, and the top 10 workers.

Regulatory and Compliance information

- SLSI Standards & Documents:
Documentation requirements specific to the Sri Lanka Standards Institute product quality, harvest procedures, and/or any environmental compliance priorities related to planning constraints.
Collection Method: Take regulatory forms and inspection reports that are current and available in records maintained presently at the premises itself or through the regulatory authorities.

The Process of Digitization and Data Collection

- Manual Record Review: All the physical written records (logbooks, ledgers, diaries) will be photographed with high-resolution cameras and systematically transcribed to enable inclusion in planning models.
- Translation Needs: Since some records could be in Sinhala or Tamil, professionally qualified translators will be contracted so that there is clear, accurate conversion to English and standardized scientific formats.
- Stakeholder Collaboration: Regular involvement in saltern ownership, management, accounting, and sales management to make clarification, confirmation and extension of data collected.
- Verification: The use of systematic information crossing (e.g. checks and balances between sales records and production records and resource assignments) again helps provide accuracy and consistency.
- Standardization: Unit size and measurements will be translated to globally accepted measures (kg, tons, LKR, USD, etc.), which contribute to interoperability and the credibility of models feeding data.

3.5. Estimated Time Frame

Table 3.5.1: Estimation Time Plan

Phase	Duration	Planned Dates (2025/26 Academic Calendar)	Key Activities
Requirement Analysis & Data Mapping	1 Month	Jul 15 – Aug 15, 2025	Identify parameters, collect sample datasets, define optimization objectives
Algorithm Design & Prototype	2 Months	Aug 16 – Oct 15, 2025	Develop initial planning algorithm, rule-based models, heuristic prototypes
Data Pipeline & API Integration	1 Month	Oct 16 – Nov 15, 2025	Set up real-time (manual/API-based) data ingestion; digitize manual records
Dashboard Development	2 Months	Nov 16, 2025 – Jan 15, 2026	Create user interface for plans, scenario testing, real-time changes
Algorithm Enhancement with AI Models	2 Months	Jan 16 – Mar 15, 2026	Add reinforcement learning/fuzzy logic, optimize models
Simulation Testing	2 Months	Mar 16 – May 15, 2026	Run historical/synthetic scenarios, collect validation metrics
Pilot Testing & Stakeholder Feedback	1 Month	May 16 – Jun 15, 2026	Deploy in selected salterns, gather feedback
Final Refinement & Documentation	1 Month	Jun 16 – Jul 15, 2026	Usability improvements, prepare deployment package, documentation

3.6. Work Breakdown Structure

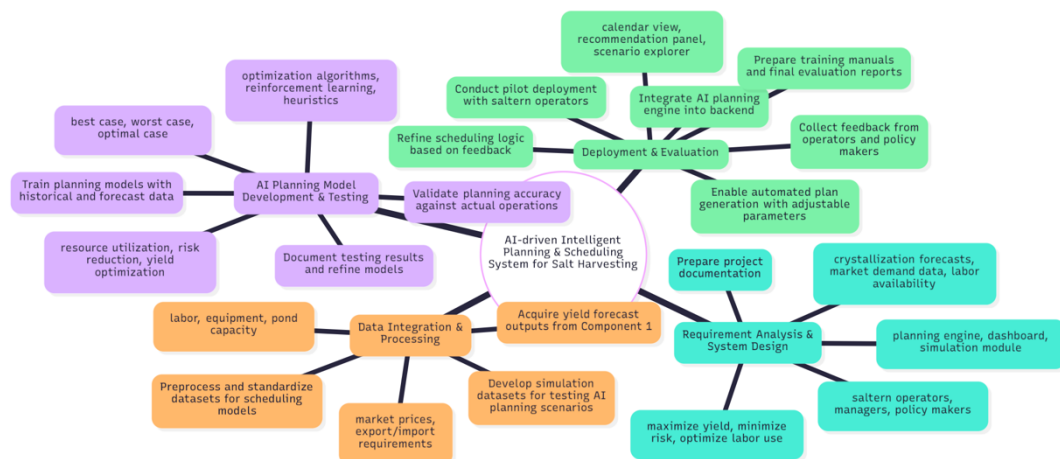


Figure 3.5.1: Work Breakdown Structure

3.7. Commercialization

Commercialization of the Intelligent Planning Engine, signals the start of what will become a commercialized solution, born as a research-applied solution, that has undergone extensive testing, gaining a stronger foothold than its predecessor and now positioned, potentially, across the salt-producing districts of Sri Lanka, and eventually across other climate-sensitive coastal areas. Because the algorithm acts as the strategic decision-making core of the Integrated Climate-Intelligent Salt Production Ecosystem, both standalone capabilities and value as a component member of an integrated system will determine commercial success.

3.7.1. Product Differentiation and Value Proposition

Intelligent Planning Engine brings a new era and style of operational planning in the Sri Lankan salt industry, with no other solution being able to provide the close connection of environmental, economic, production, and regulatory data being closely connected to real-time adaptive decision support.

Key Differentiators:

- **Multi-Parameter Optimization:** Reconciles climatic predictions, market trends, and cost control, and quality warranting to generate a highly customizable and value-added operational strategy.
- **Dynamic Adaptation: In Real-Time:** Updates recommendations according to incoming data on a continuous basis, allowing responsive, risk-sensitive operations in a period of climatic and market volatility.
- **Stakeholder-Oriented Customization:** Enables producers to customize strategy to the business objectives (e.g. export thrust versus domestic volume, cost management, quality standards).
- **Scalability and Modularity:** Can be deployed per single-location or multi-location producers; can simply be expanded and connected to the ecosystem.
- **Robustness:** Flawless interoperability with other core modules (crystallization forecasting, quality control and waste valorization).
- **Value Proposition:** Enables salt producers to make decisions about their operations based on real data, resilient, and profitable to reduce risk, align output to market demand, and pursue long-term profitability in a highly volatile climate.

3.7.2. Revenue Model

A flexible Software-as-a-Service (SaaS) subscription model will be used in commercial deployment of:

Subscription Tiers

- Basic: Low monthly costs; comes with manual data entry, basic planning analytics.
- Professional: Improved real-time data integration capability, increased customization, and scenario analytics.
- Enterprise: Integrated package that covers ecosystem, predictive analysis, custom support and sensor connections (optional).

Other sources of income

- Special Set-up Charges: Supplies onboarding, migration, and customization of new clients.

Optional Add-Ons:

- Provides access to sophisticated analytics modules, APIs in ERPs, consulting and training packages to suit the unique demands of a client.

3.7.3. Scalability and Market Penetration

The solution can be used by both large-scale salt producers, and small/medium salterns in hubs (Hambantota, Puttalam, Jaffna), assisting collaboration and competitiveness in the chain.

Expansion (regional and global): After local validation, the algorithm can be tuned towards markets in India, Bangladesh and Southeast Asia- all of which have comparable coastal climatic risks, and production arrangements.

Scalability Features:

- Remote access and maintenance Cloud-native deployment.
- Upgrades based on modules, that would allow the kind of permanent evolution of features and efficiency.
- Multi-language interface so it can be adopted regionally more widely.

3.7.4. Marketing Strategy

A multi-faceted approach will be taken to establish credibility and market adoption:

- Industry Partnerships: Work with national agencies (Puttalam Salt Society) to seek endorsement, pilot sites.
- Field Demonstrations: Write up and share actual world success stories of partner salterns who have used the system.
- Training & Workshops: Conduct operator, managerial and stakeholder hands-on training to achieve ease of implementation and confidence of the users.
- Digital Outreach: Conduct special programs on efficiency, profitability and sustainability enhancement through social media, webinars, and trade magazines.

4. PROJECT REQUIREMENTS

4.1. Functional Requirements

Acquisition & Integration of data

FR1: The system will obtain live environmental data (weather, climate, etc.) through built-in APIs.

FR2: Planning workflow communications will automatically ingest Component 1 forecasts of production and crystallizations and match them to planning materials.

FR3: Prices, demand, and trends information will be retrieved on the market via reliable online databases and governmental portals. Map & Store of Parameters

FR4: Input parameters acquired will be centralized securely in a database so that they are consistent and can be accessed quickly.

FR5: Parameters will automatically be refreshed/updated to indicate the most up-to-date data.

Plan Generation

FR6: The IPA will give optimized operational plans balancing environmental, market, resource, and regulatory constraints with advanced algorithms.

FR7: Users will have a chance to choose the type of specific optimization that they desire, e.g., to maximize profit, quality, or sustainability. Plan Customization

FR8: The business objectives (home/analysis focus, risk acceptance) can be determined by shareholders/managers.

FR9: The user preferences will also be considered: the algorithmic advising and plan outputs will dynamically adapt.

Dashboard & Visualization

FR10: New operational plans will be generated in the dashboard with visually clear representations, KPIs, and summary metrics.

FR11: The updated parameters are visible to the user, and the influence of the parameters on the operational outcomes is reflected at once.

FR12: There will be a scenario testing tool that will enable the user to test changes to the conditions and other action plans.

Alerts & Notifications

FR13: Automated messages containing information about changes in some important parameters will notify the users of the necessity to update their plan.

Testing & Validation

FR14: Scenario and simulation testing will be supported by the system based on historical data to validate planning logic and results.

FR15: Every piece of feedback, provided by a user will be recorded and utilized in subsequent iterations and as the system develops.

4.2. Non-Functional Requirements

Performance: Recalculate plans and present them within five seconds of entry of any new or updated data.

Scalability: The system should be in a position to undertake simultaneous planning of a series of salterns without impairing performance.

Security: The transmission of all data and its storage will be performed with modern encryption protocols (HTTPS/TLS).

Usability: The interface needs to be user-friendly and easy to use with little training in order to achieve successful utilization.

Availability: To schedule the production daily, the system is expected to retain 99 percent uptime.

Maintainability: Modularity of the platform is expected to support easy improvement, debugging, and the incorporation of features.

4.3. Test Cases

Table 4.3.1: Test Cases

Test Case ID	Description	Input	Expected Output	Pass Criteria
TC01	Validate market data API connection	API key, request params	Valid JSON response with market data	200 response, correct data returned
TC02	Generate plan with profit goal	All relevant planning data	Optimized plan maximizing projected profits	Plan yields $\geq$ baseline profit
TC03	Respond to sudden price drop	Updated price input	Adjusted plan reflecting diminished value	Update within 5 seconds
TC04	Apply user export preference	User selects export focus	Majority (>80%) export-grade salt in plan	Plan matches preference
TC05	Display plan on dashboard	Plan object	UI presents plan, charts, and KPIs	Full visibility, correct data

4.4. Wireframe

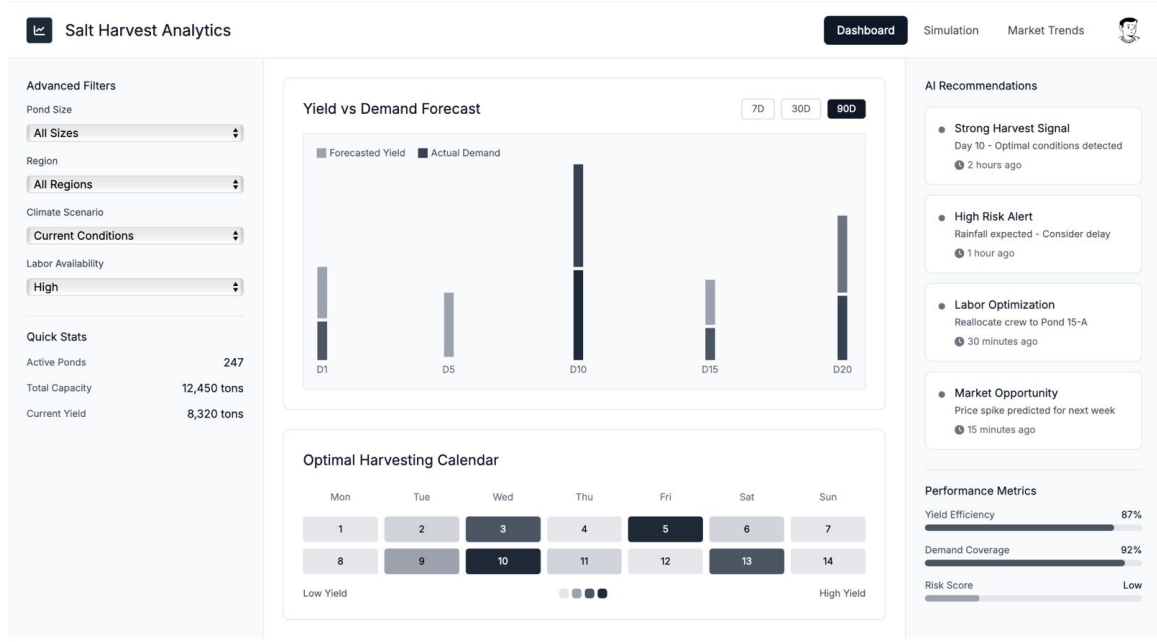


Figure 4.4.1: Wireframe For Intelligent Planning Dashboard

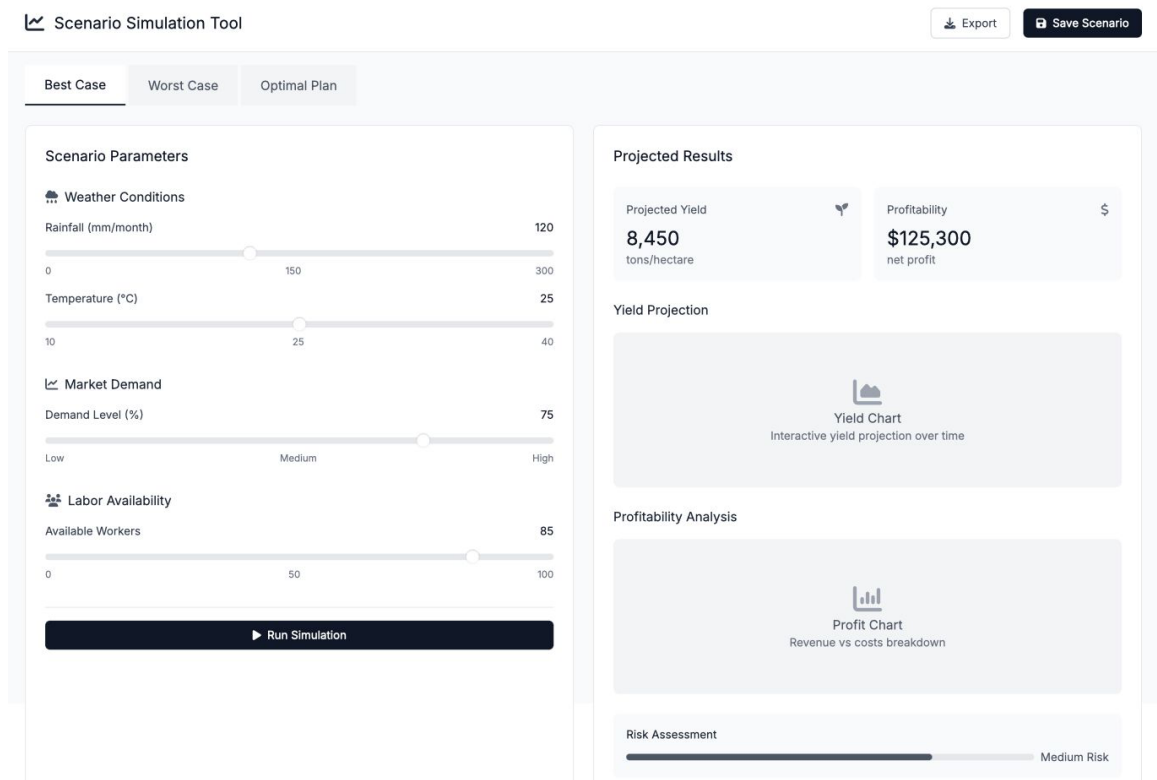


Figure 4.4.2: Wireframe for Scenario Simulation Dashboard

5. BUDGET AND BUDGET JUSTIFICATION

Expense	Cost (LKR)	Net Total	Justification
Personal Costs			
Travelling	36,000	36,000	3000 x 12 covering site visits to Puttalam and supervisor meetings.
Software And Service			
Domain Purchase	1,000		4000 / 4 mem
Host and deploy	17,500		AWS EC2 Instance (200\$) 70,000/4
Perplexity Pro	18,000		6000 x 12 / 4
Premium Api	12,500		For data gathering (e.g.: weather) 50,000 / 4
Google Collab Pro	36,000		For model training (10\$ = 3000 per month)
Cloud Storage	20,000	105,000	Additional storage if needed
Grand Total		141,000	

6. REFERENCES

- [1] T. Morning.lk, "Sri Lanka to import salt starting tomorrow," *Sri Lanka to import salt starting tomorrow*, 2 January 2025.
- [2] IPCC, "CLIMATE CHANGE 2023 Synthesis Report," 2023. [Online]. Available: [https://www.perplexity.ai/search/i-am-writin-my-reserch-project-isGqwfZ4TluuqPtWIsS6hA?58=r#:~:text=PDF\]%20CLIMATE-,CHANGE,-2023.](https://www.perplexity.ai/search/i-am-writin-my-reserch-project-isGqwfZ4TluuqPtWIsS6hA?58=r#:~:text=PDF]%20CLIMATE-,CHANGE,-2023.)
- [3] R. Diwyanjalee, "How Climate Change is Reshaping Sri Lanka's Salt Industry," 01 January 2025. [Online]. Available: <https://climatefactchecks.org/how-climate-change-is-reshaping-sri-lankas-salt-industry/>.
- [4] A. S. B. & L. C. Jones, "Solar evaporation and salt production: Environmental considerations.," *Renewable Energy*, 2018.
- [5] H. & S. D. Perera, "Salt production in Sri Lanka: Present status and challenges.," *Economic Botany*, 2022.
- [6] W. e. a. Fernando, "Impact of climate variability on Sri Lankan coastal salt production.," *Journal of Environmental Management*, 2021.
- [7] L. & R. S. Wijesekara, "Challenges in salt production planning in Sri Lanka.," *Journal of Salt Science and Technology*, 2020.
- [8] J. e. a. Zhang, "AI-powered crop planning for climate-smart agriculture," *Agricultural Systems*, 2020.
- [9] P. e. a. Duarte, "Predictive analytics for sustainable fisheries management.," *Marine Policy*.
- [10] Y. & Z. H. Wang, "Reinforcement learning for resource scheduling in manufacturing," *IEEE Transactions on Industrial Informatics*.
- [11] FAO., "International standards for salt purity and quality. Food and Agriculture Organization".
- [12] S. L. Ministry of Trade, "Salt industry regulatory standards report. Government of Sri Lanka," 2023.

- [13] M. I. W. K. Y. H. Kasnir Muhammad, "Feasibility Study and Factors Influencing Salt Industry Production," Asian Journal of Social Science and Management Technology, Indonesia, 2022.
- [14] A. C. P. Hub, "alibabaCloud," SaltBae: AI-Powered Salt Production Prediction, 15 November 2021. [Online]. Available: https://www.alibabacloud.com/blog/saltbae-ai-powered-salt-production-prediction_598242.
- [15] N. S. C. o. S. Lanka, "Annual Salt Production Reports.," [Online]. Available: <http://www.nascorp.lk/>.
- [16] O. API., "Real-time Weather Data for Developers.," [Online]. Available: <https://openweathermap.org/api>.
- [17] S. L. S. I. (SLSI), "Specifications for Food Grade Salt.," [Online]. Available: <https://www.slsi.lk/>.
- [18] S. & B. A. Ghosh, "Multi-objective optimization in supply chain management," *A review and future research directions*, 2020.
- [19] S. L. P. Authority, "Export Logistics and Salt Shipment Procedures.," [Online]. Available: <http://www.slpa.lk/>.
- [20] C. M. A. S. & S. R. Duarte, "Sustainable fisheries management in a changing climate.," *Sustainable fisheries management in a changing climate.*, 2019.
- [21] v. p. a. H. o. B. S. T. W. C. a. I. B. By Mark Gildersleeve, "IBM Watson Decision Platform for Agriculture," IBM, 2024. [Online]. Available: <https://newsroom.ibm.com/IBM-watson?item=30660>.

7. APPENDICES